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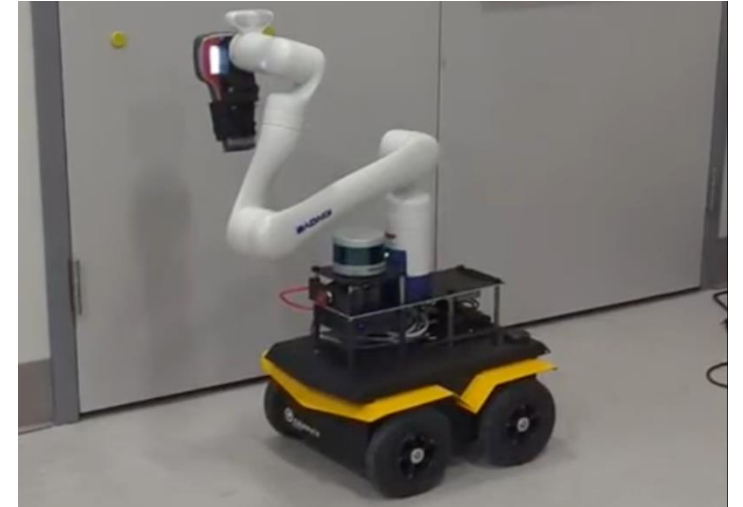
Data Efficient Localization and Mapping for Distributed Multi-Robot Teams in the Field

Yewei Huang

The Robust Field Autonomy Lab
Department of Mechanical Engineering
Stevens Institute of Technology

Motivation: Why Multi-Robot Teams?

- Multi-robot teams can be more efficient than a single robot in **time-sensitive** or **collaborative** tasks:
 - **Disaster Rescue** – Faster search and coordinated response.
 - **Exploring Hazardous Environments** – Autonomous mapping, real-time monitoring.
 - **Agricultural & Construction Services** – Efficient large-scale operation, task division.



Mobile Manipulation Platform for Autonomous Indoor Inspections in Low-Clearance Areas.



UUV from Bluefin Robotics.



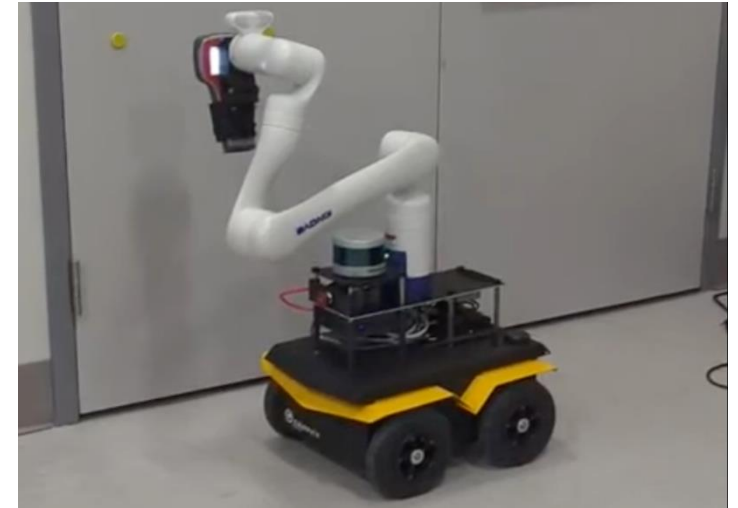
Jackal UGV with LiDAR and IMU on the Campus.



Customized Built BlueROV2 Surveying the Water Tank.

Motivation: Perception in Robot Teams

- Perception plays a crucial role in:
 - **Ensuring Safe Navigation** – Avoiding obstacles, adapting to dynamic environments.
 - **Supporting Future Task Allocation** – Providing map-based decision-making for team coordination.
- **Core Perception Module for Unknown Environments:**
 - **Simultaneous Localization and Mapping (SLAM).**



Mobile Manipulation Platform for Autonomous Indoor Inspections in Low-Clearance Areas.



UUV from Bluefin Robotics.



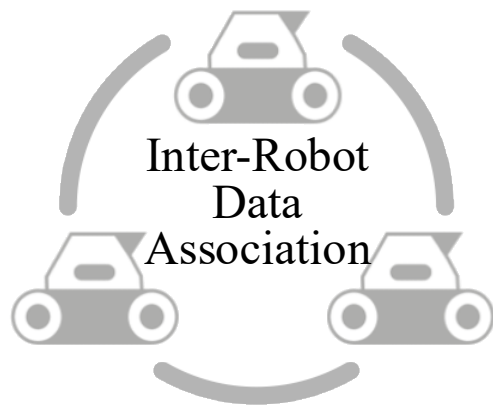
Jackal UGV with LiDAR and IMU on the Campus.



Customized Built BlueROV2 Surveying the Water Tank.

Motivation: Challenges for Multi-Robot SLAM

- Simultaneous Localization and Mapping (SLAM):
 - A robotics technique that enables a robot to **map an unknown environment** while simultaneously **tracking its own location** within that map.
- Multi-Robot SLAM:
 - Multiple robots collaboratively **map the environment** and **localize themselves** within it.
- Main Challenges in **Multi-Robot SLAM**:
 - **Limited Communication Bandwidth** – Restricts information sharing and coordination.
 - **Accurate Priors for Map Merging** – Crucial for reliable map data association.
 - **Map Consistency** – Ensures a unified and coherent map across all robots.
 - **Robustness to Failures** – Maintains accuracy despite robot or communication failures.

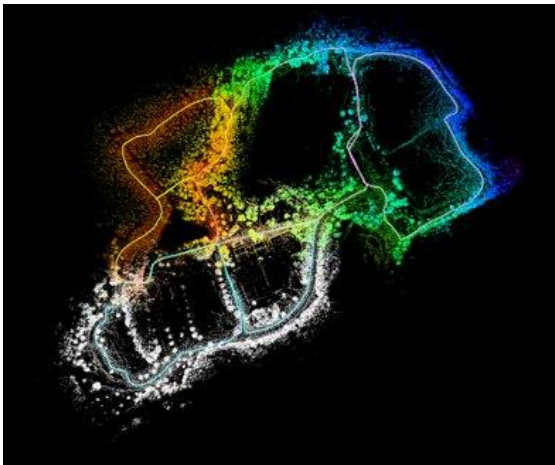


Merged Map

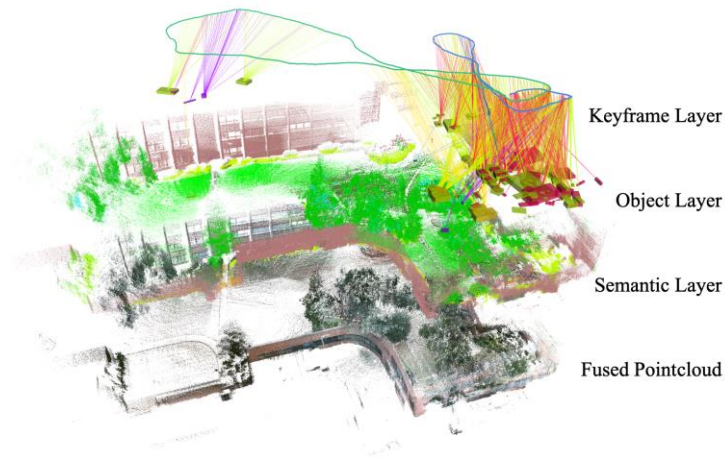
How to perform multi-robot SLAM **efficiently** and **effectively** when we only have limited communication bandwidth?

Our Approach

- **DiSCo-SLAM**: A distributed multi-robot SLAM framework using **compact LiDAR descriptor** intended for real-time use with 3D LiDAR, and a **two-stage** global-local factor graph **optimization**.
- **SGM-SLAM**: The first approach to solving the geometric prior for multi-robot data association only with **scene graph matching**, utilizing limited semantic and geometric information.
- **DRACo-SLAM2**: The Group-wise Consistent Measurement Set Maximization (GCM) algorithm for **robust loop closure selection**, targeting cases where multiple loop closures exhibit similar registration errors.



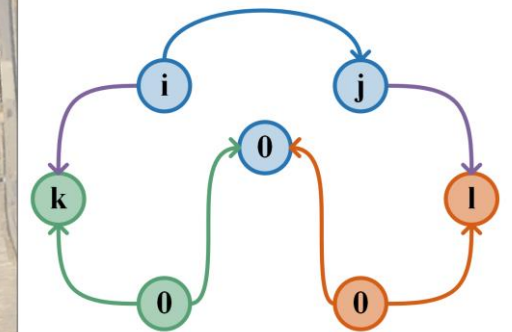
SLAM map generated by multi-robot DiSCo-SLAM framework.



Multi-layer scene graph merged by graph matching.



Multi-robot sonar SLAM results with GCM-based loop closure selection.





DiSCo-SLAM

- Using Scan Context descriptor^[1] in multi-robot LiDAR SLAM for **data-efficient communication**.
- A two-stage global-local optimization for each robot:
 - global optimization: high quality **initial guess**;
 - local optimization: stable and accurate **local pose graph**.

[1] Kim, Giseop, and Ayoung Kim. "Scan context: Egocentric spatial descriptor for place recognition within 3d point cloud map." 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2018.



Our Jackal UGV with LiDAR, IMU and GPS.

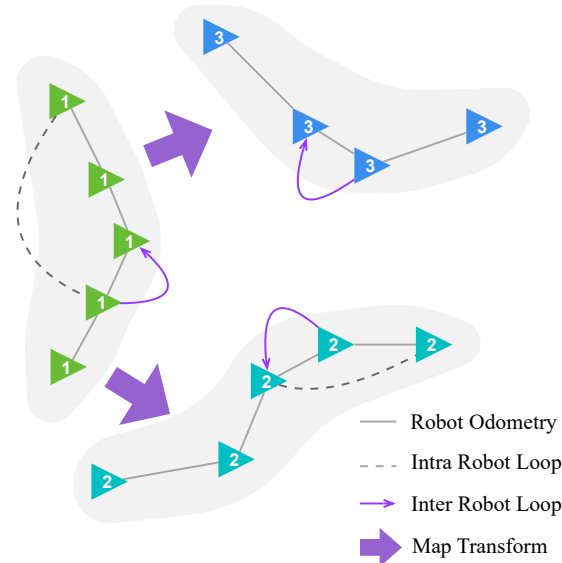
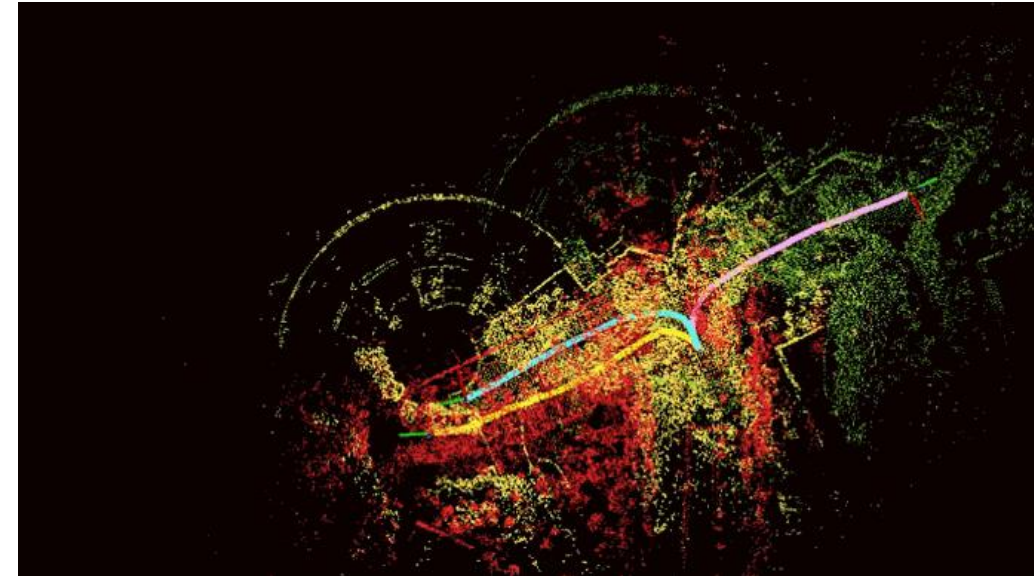
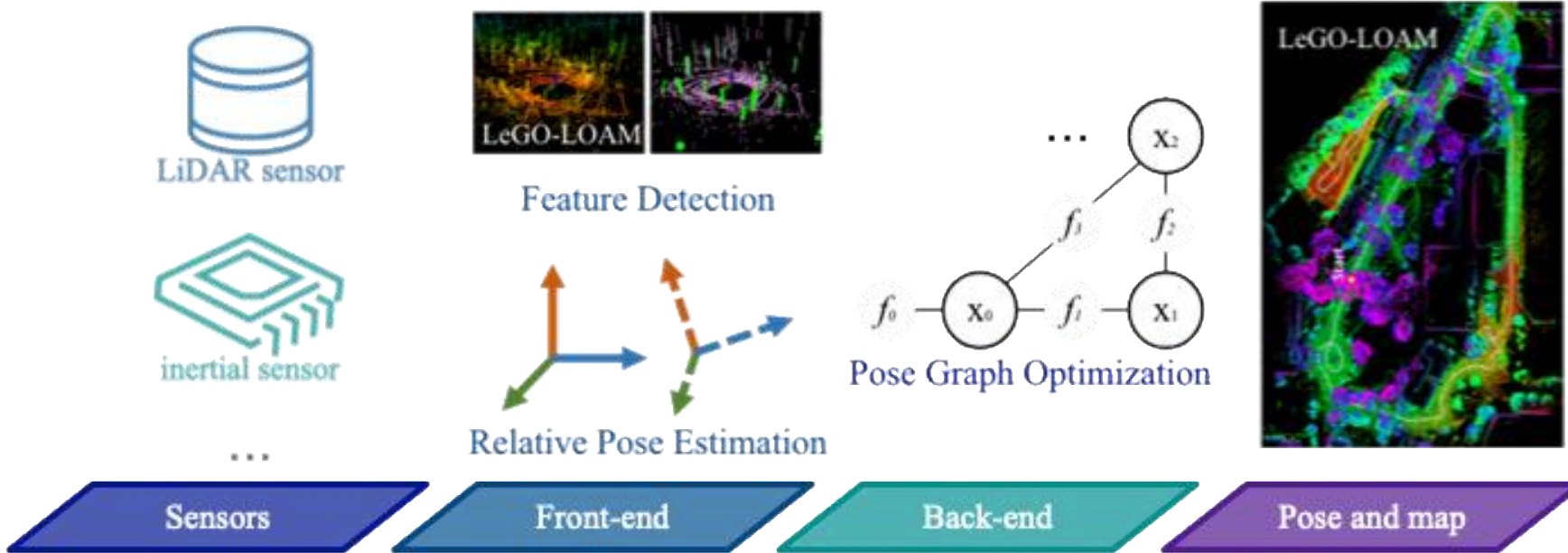


Illustration of the proposed method, where a global coordinate transformation graph (purple) and local pose graph (gray) are optimized by each robot.



DiSCo-SLAM Performance on the Graco Dataset.

Single Robot SLAM



In Simultaneous localization and mapping (**SLAM**), a mobile robot maps an **un-known** environment, while using **relative** measurements of that environment as the basis for **localizing** itself.

Given:

- Set of robot states: $\mathcal{X}$
- Corresponding observations: $\mathcal{Z}$

Define:

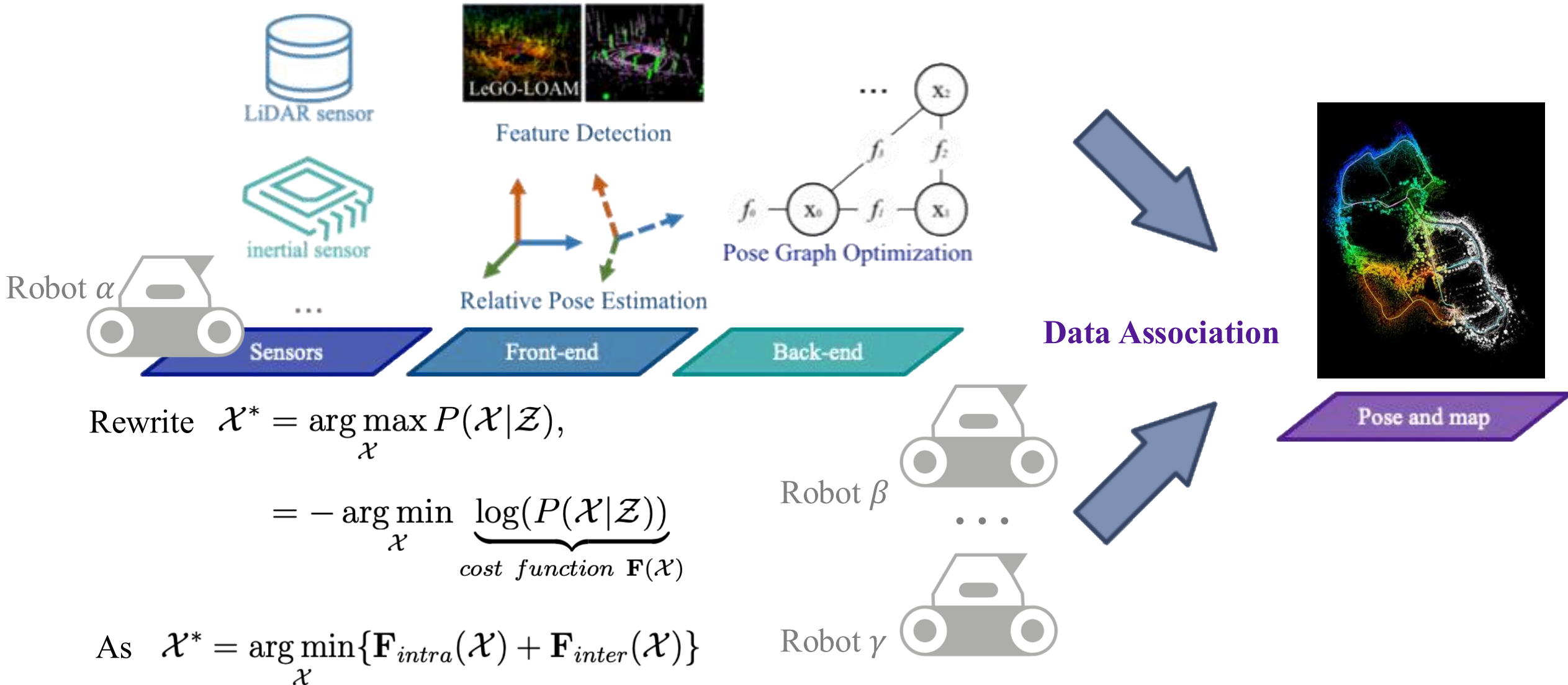
- The SLAM problem is then represented as a maximum

a posteriori (**MAP**) problem: $\mathcal{X}^* = \arg \max_{\mathcal{X}} P(\mathcal{X}|\mathcal{Z}),$

$$= - \arg \min_{\mathcal{X}} \underbrace{\log(P(\mathcal{X}|\mathcal{Z}))}_{\text{cost function } \mathbf{F}(\mathcal{X})}$$



Distributed Multi - robot SLAM





Distributed Multi - robot SLAM

- Rewrite $\mathcal{X}^* = \arg \min_{\mathcal{X}} \{\mathbf{F}_{intra}(\mathcal{X}) + \mathbf{F}_{inter}(\mathcal{X})\}$ to **distribute the task to local robots**:

$$\mathbf{F}_{intra}(\mathcal{X}) = \sum_{\alpha \in \mathcal{N}} \sum_{\langle i,j \rangle \in \mathcal{C}_{\alpha}} \mathbf{F}_{ij} \quad \mathbf{F}_{inter}(\mathcal{X}) = \sum_{\alpha, \beta \in \mathcal{N}, \alpha \neq \beta} \sum_{\langle i,j \rangle \in \mathcal{C}_{\alpha\beta}} \mathbf{F}_{ij}$$

- $\mathcal{C}_{\alpha}$: the set of **intra-robot** constraints among poses of robot
- $\mathcal{C}_{\alpha\beta}$: the **inter-robot** constraints pertaining to poses of robots α and β
- For local robot α :

$$\mathbb{X}_{\alpha}^* = \arg \min_{\mathbb{X}_{\alpha}} \left(\overbrace{\sum_{\langle i,j \rangle \in \mathcal{C}_{\alpha}} \mathbf{F}_{ij}}^{\mathbf{F}_{intra}} + \overbrace{\sum_{\beta \in \mathcal{N}, \alpha \neq \beta} \sum_{\langle i,j \rangle \in \mathcal{C}_{\alpha\beta}} \mathbf{F}_{ij}}^{\mathbf{F}_{inter}} \right)$$

$$\mathbb{X}_{\alpha} = \mathcal{X}_{\alpha} \cup \left\{ \mathbf{x}_j \left| \begin{array}{l} \text{Pose of robot } \beta \text{ at timestamp } j \\ \mathbf{x}_j \in \mathcal{X}_{\beta}, \langle i, j \rangle \in \mathcal{C}_{\alpha\beta}, \\ \forall \beta \in \mathcal{N}, \beta \neq \alpha. \end{array} \right. \right\}$$



Two Stage Global Local Optimization

- Global Optimization:

- Separator pose $\langle \alpha_i, \beta_j \rangle \in \mathcal{C}_{\alpha\beta}$, with $\mathbf{x}_{\alpha_i} \in \mathcal{X}_\alpha$, $\mathbf{x}_{\beta_j} \in \mathcal{X}_\beta$
- Define local robot frame for robot α as: $\mathcal{F}_\alpha$

$${}^\beta \mathbf{x}_{\alpha_i} = \mathbf{T}_{\beta\alpha} \cdot {}^\alpha \mathbf{x}_{\alpha_i} = {}^\beta \mathbf{x}_{\beta_j} \cdot {}^\beta \mathbf{z}_{\beta_j\alpha_i}$$

$$\mathbf{T}_{\beta\alpha} = {}^\beta \mathbf{x}_{\beta_j} \cdot {}^\beta \mathbf{z}_{\beta_j\alpha_i} \cdot ({}^\alpha \mathbf{x}_{\alpha_i})^T.$$

- ${}^\beta \mathbf{x}_{\alpha_i}$: Pose at time i of robot α in robot β 's local frame
- $\mathbf{T}_{\beta\alpha}$: Transformation matrix from ${}^\alpha \mathbf{x}_{\alpha_i}$ to ${}^\beta \mathbf{x}_{\alpha_i}$
- Define the set of transformations from any robot frame to the global frame:

$$\mathbb{T} = \{\mathbf{T}_{g\alpha} \mid \forall \alpha \in \mathcal{N}, \alpha \neq g\}$$

- Global** Optimization: $\mathbb{T}^* = \arg \min_{\mathbb{T}} \sum_{\alpha, \beta \in \mathcal{N}} \mathbf{e}_{\beta\alpha}^T \Omega_{\beta\alpha} \mathbf{e}_{\beta\alpha}$

- Negative log-likelihood function: $\mathbf{e}_{\beta\alpha}(\mathbf{T}_{g\beta}, \mathbf{T}_{g\alpha}) = \mathbf{T}_{\beta\alpha}^{(i)} - \hat{\mathbf{T}}_{\beta\alpha}^{(i)}(\mathbf{T}_{g\beta}, \mathbf{T}_{g\alpha})$

- Local** Optimization $\mathcal{X}^* = \arg \min \{\mathbf{F}_{intra}(\mathcal{X}) + \mathbf{F}_{inter}(\mathcal{X})\}$

$$\mathbf{F}_{inter} = \sum_{\substack{\alpha, \beta, \gamma \in \mathcal{N}, \\ \alpha \neq \beta, \alpha \neq \gamma}} \sum_{\substack{\langle \alpha_i, \beta_j \rangle \in \mathcal{C}_{\alpha\beta}, \\ \langle \alpha_k, \gamma_l \rangle \in \mathcal{C}_{\alpha\gamma}}} \mathbf{e}_{vir}^T \Omega_{vir} \mathbf{e}_{vir}$$

$$\mathbf{e}_{vir} = \mathbf{z}_{\alpha_i\beta_j} \cdot {}^\alpha \mathbf{z}_{\beta_j\gamma_l} \cdot (\mathbf{z}_{\alpha_k\gamma_l})^T - \hat{\mathbf{z}}_{\alpha_i\alpha_k}(\mathbf{x}_{\alpha_i}, \mathbf{x}_{\alpha_k})$$

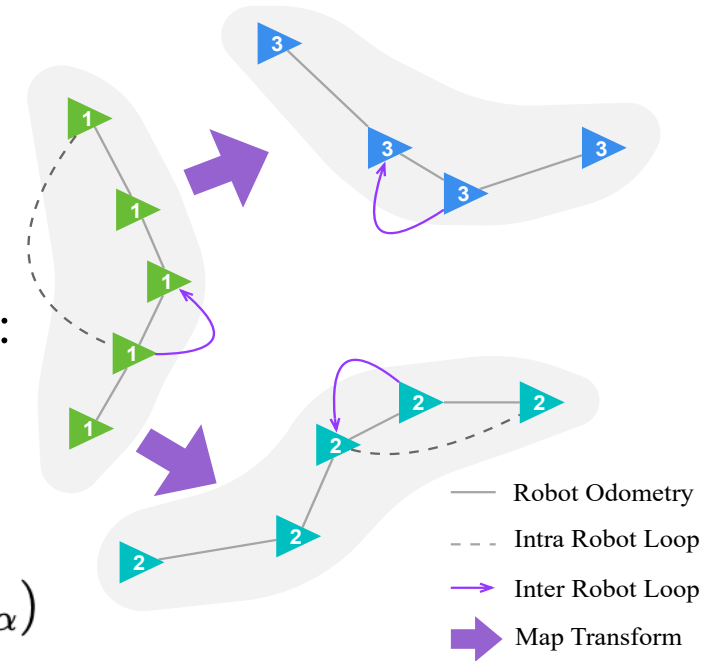
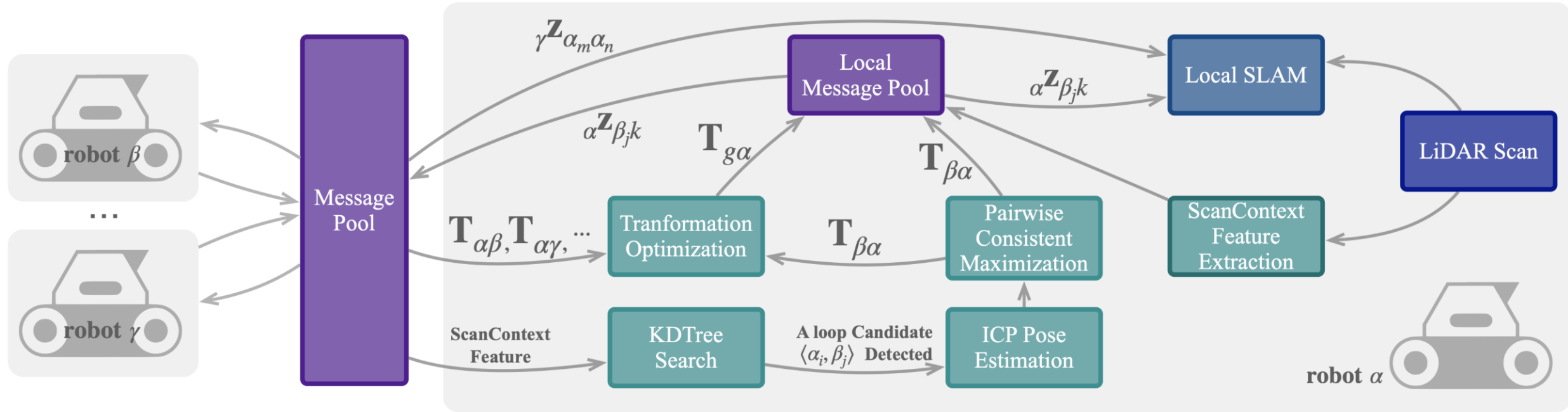


Illustration of the proposed method, where a global coordinate transformation graph (purple) and local pose graph (gray) are optimized by each robot.



An overview of the architecture of DiSCo-SLAM





DiSCo-SLAM with three different configurations on the park dataset

- The data is processed in real-time;
- The video speed is 80x.

Full global and local optimization

Local and global optimization
without inter-robot constraints

Only global transformation
optimization

■ Jackal 0
■ Jackal 1
■ Jackal 2



DiSCo-SLAM with three different configurations on the park dataset

- The data is processed in real-time;
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Full global and local optimization



■ Jackal 0
■ Jackal 1
■ Jackal 2

Local and global optimization
without inter-robot constraints



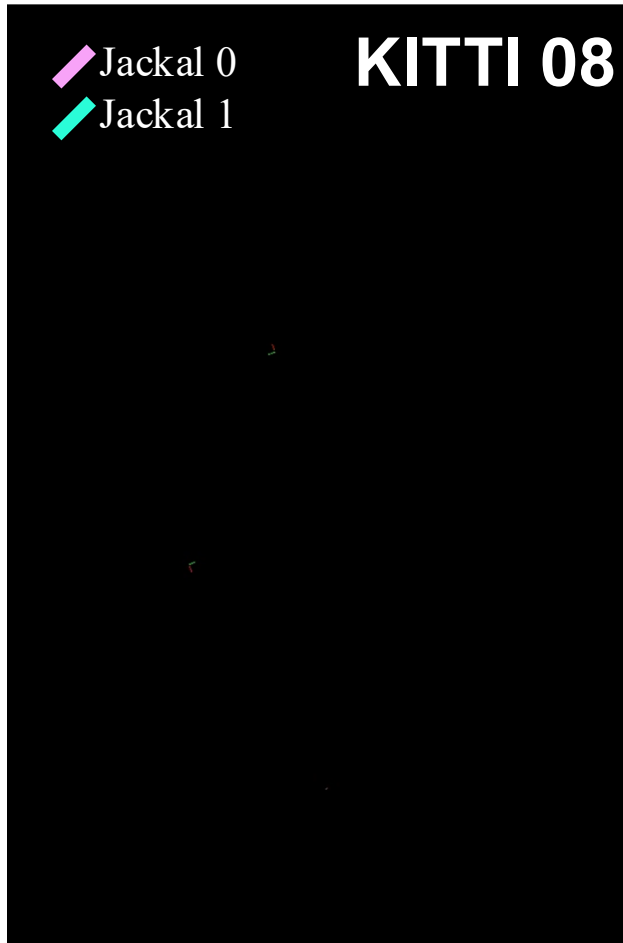
Only global transformation
optimization



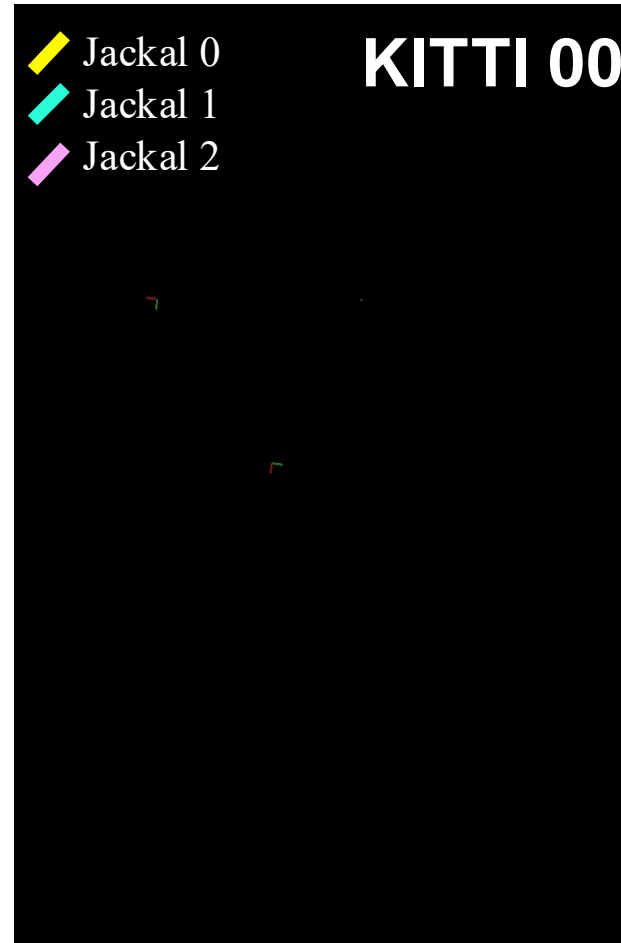


DiSCo-SLAM on KITTI 00 & KITTI 08 & Stevens Dataset

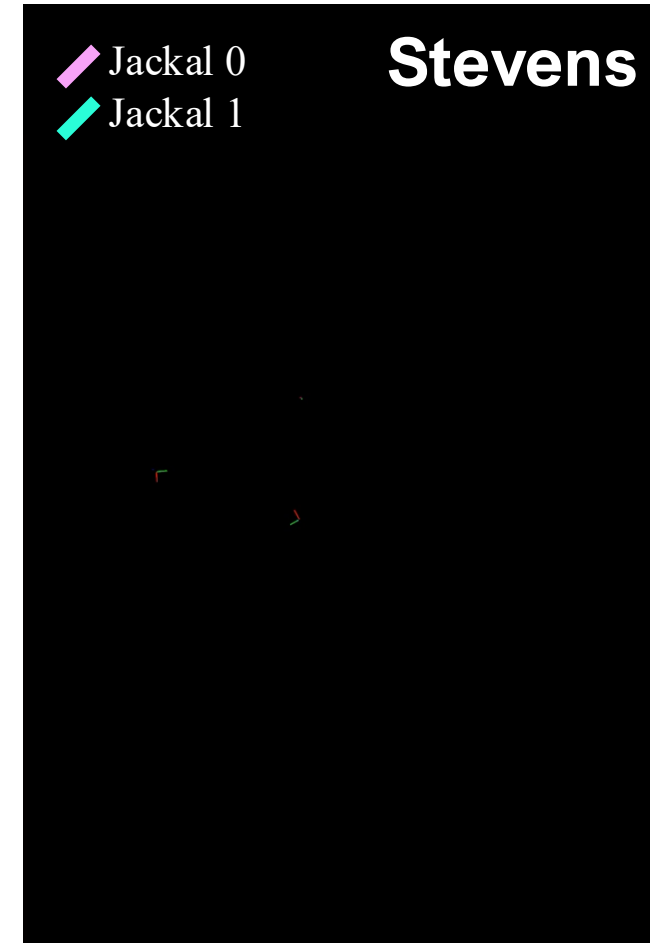
The data is processed in
real-time;
The video speed is 20x.



The data is **not** processed in
real-time;
The video speed is 15x.

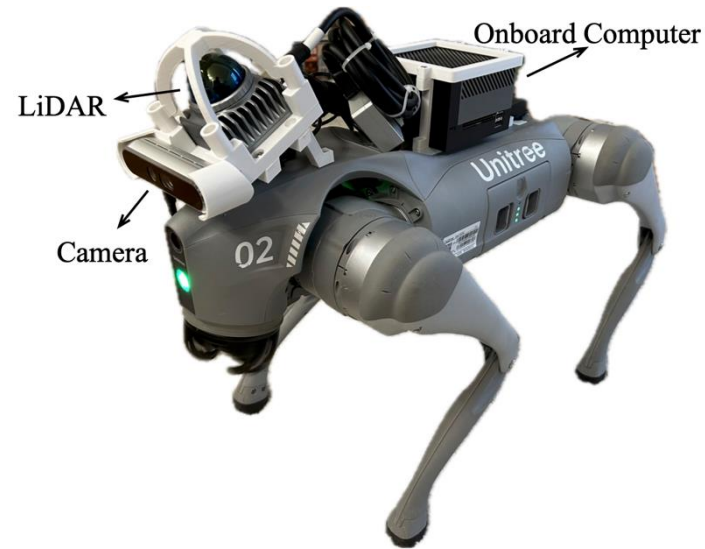


The data is processed in
real-time;
The video speed is 80x.

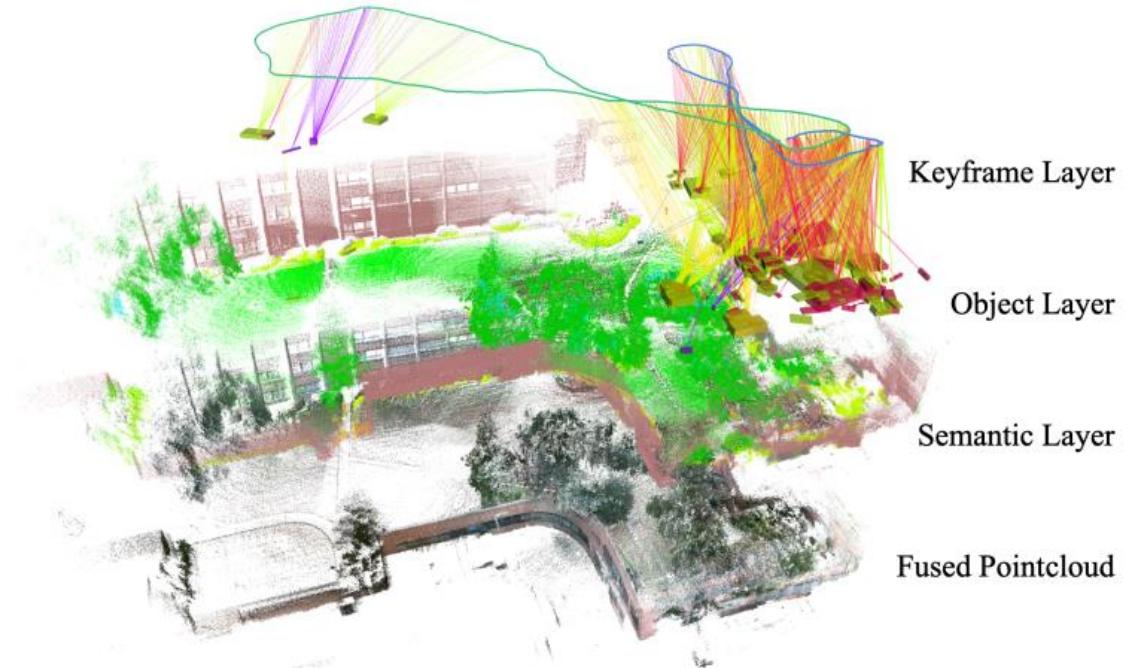


Motivation: Scene Graph SLAM

- With the recent development of Large Language Models (LLMs), there is a growing need to understand the environment at a **semantic level** to support tasks such as motion planning and human-robot interaction.
- Lidar-equipped mobile robots are typically range and **bandwidth limited** and must be selective in exchanging information.



Unitree Go2 EDU equipped with camera and LiDAR.



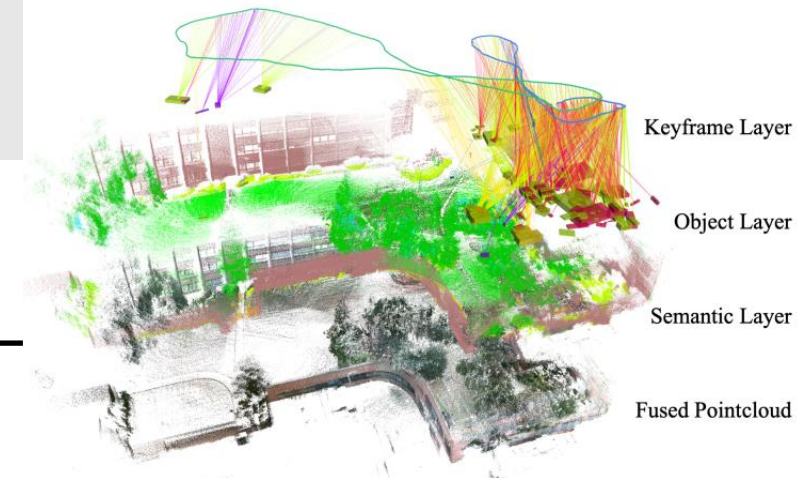
Scene graph constructed by scene graph matching.

Motivation: Why Scene Graph Map?

Feature	Scene Graph Maps	Point Cloud Maps
Data Size	Lightweight , efficient for transmission	High data volume, resource-intensive
Structure	Hierarchical , intuitive for merging	Flat, requires extra data for merging
Human Interpretation	Straightforward and easy to interact with	Requires additional processing for understanding
Semantic Information	Rich, inherently structured	Limited, needs extra processing

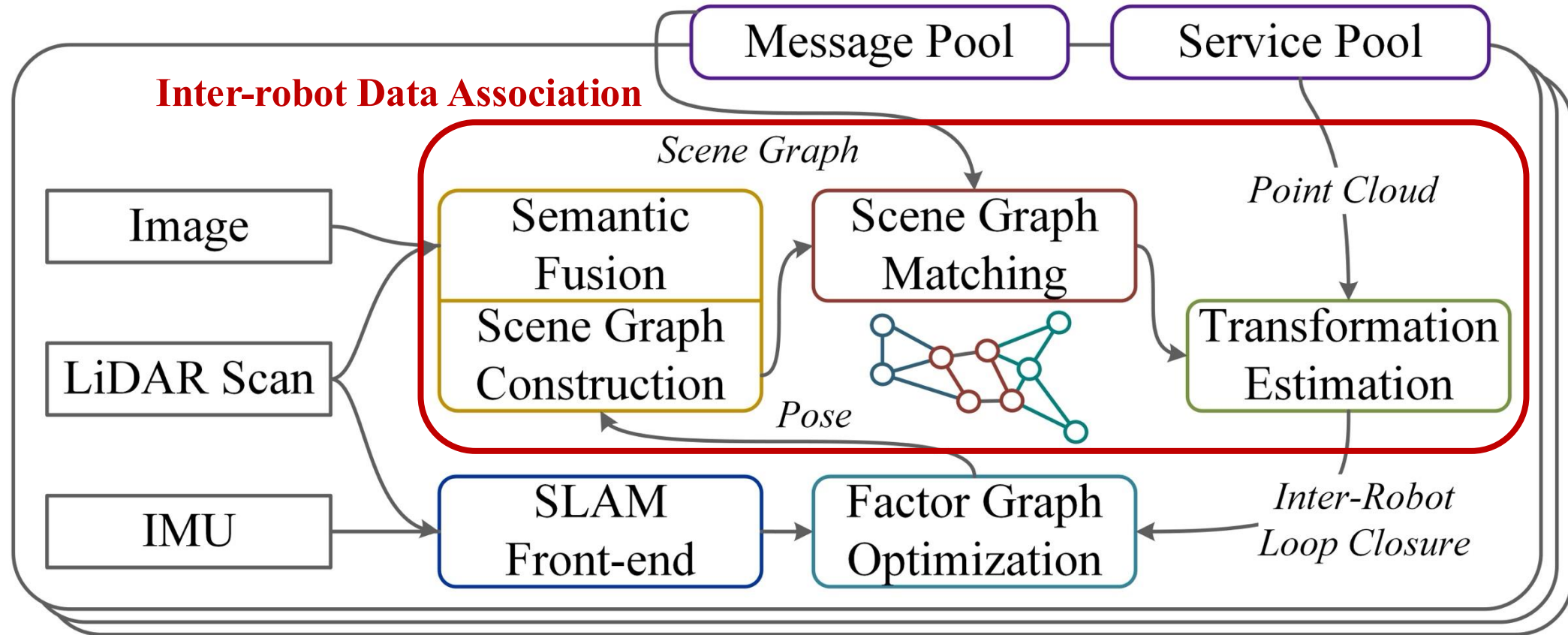


Point cloud map from DiSCo-SLAM.



Scene graph map from Scene Graph SLAM.

An overview of the architecture of Scene Graph SLAM



Single Robot Scene Graph Construction

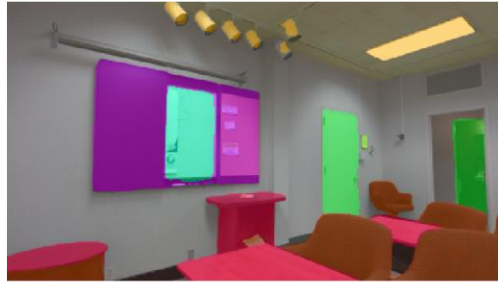
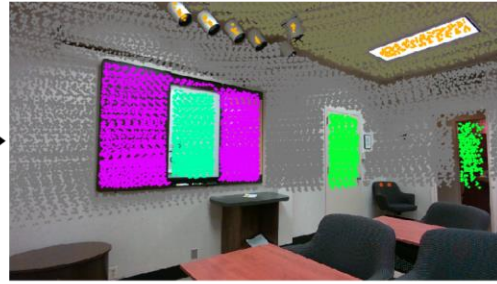
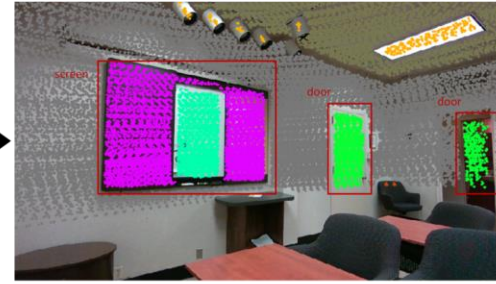


Image Segmentation



LiDAR Camera Point-cloud Association



Object Clustering

Intra-Robot Data Association

$$c_{jk} = \|p_j - p_k\|_2$$

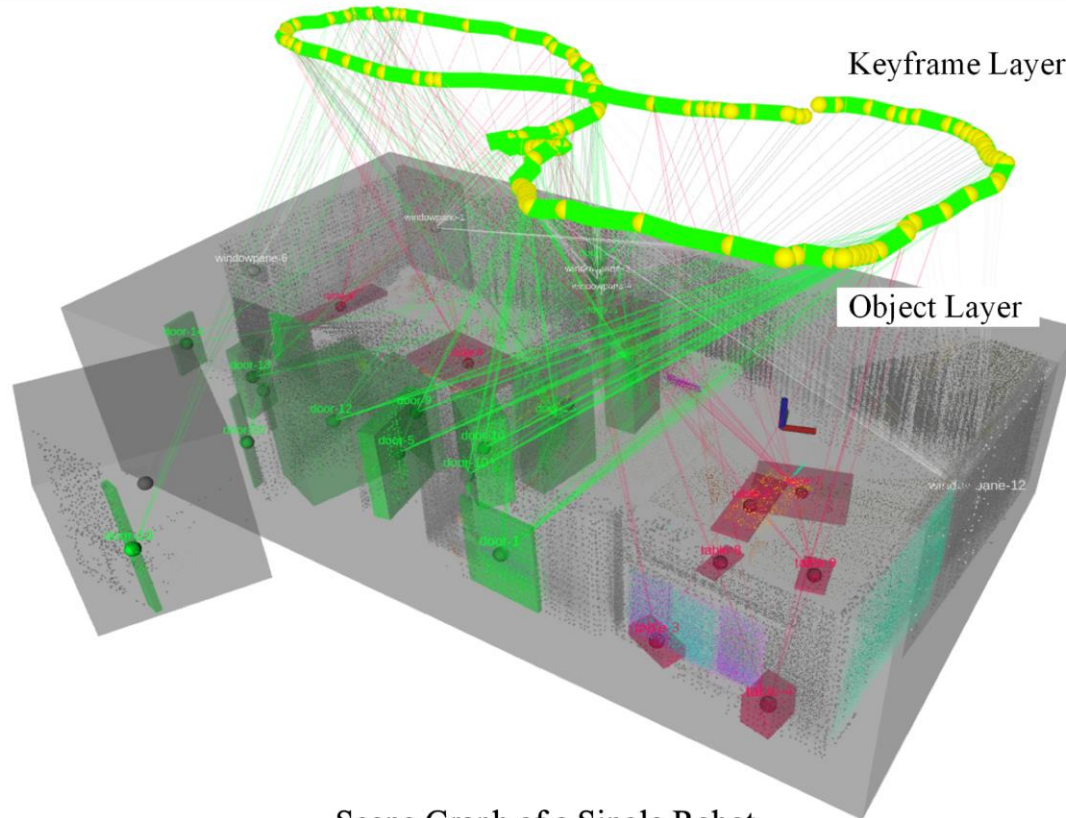
Euclidean distance between the centers of two objects

Linear Assignment Problem(LAP) :

$$A_n^* = \arg \min_{A_n} \sum_{v_j \in \mathcal{V}_l^{(t)}} \sum_{v_k \in \mathcal{V}_l^{(t-1)}} c_{jk} \cdot a_{jk};$$

$$\forall v_k \in \mathcal{V}_l^{(t-1)}, \sum_{v_j \in \mathcal{V}_l^{(t)}} a_{jk} = 1;$$

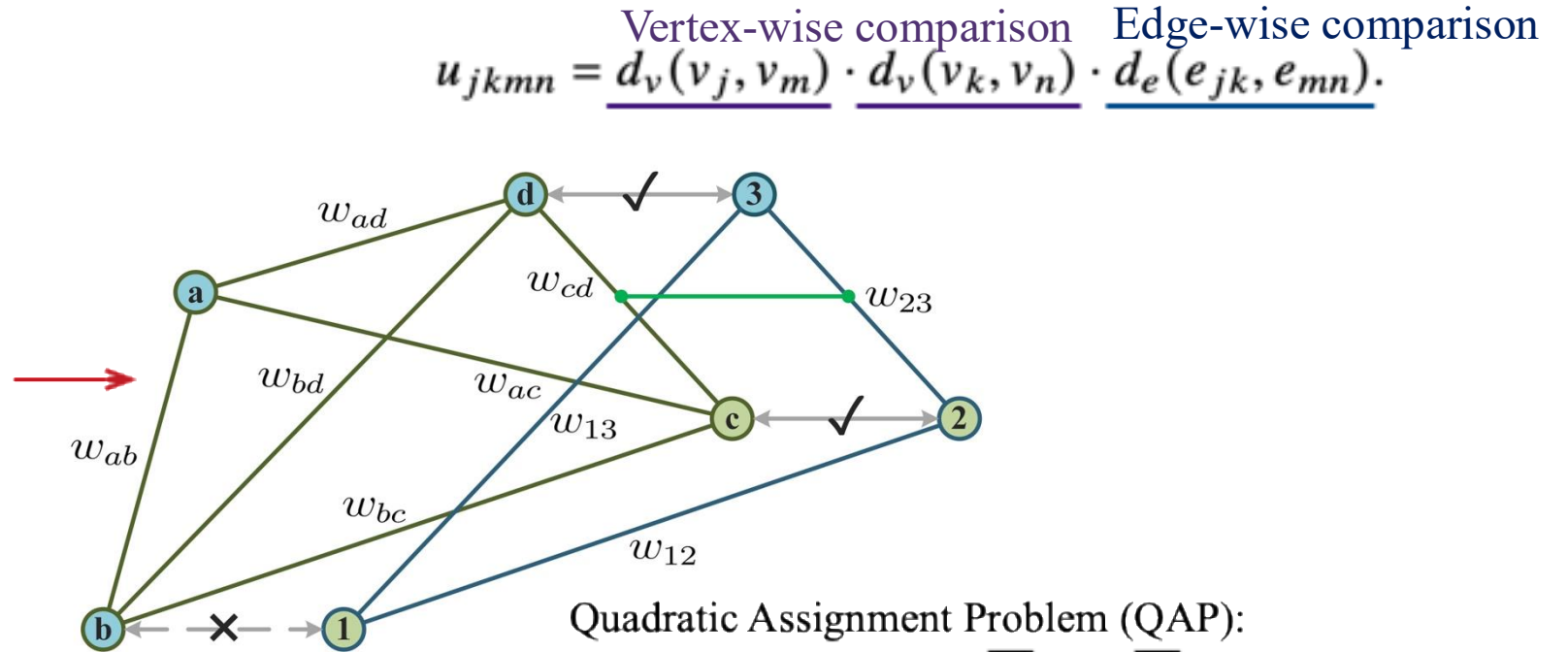
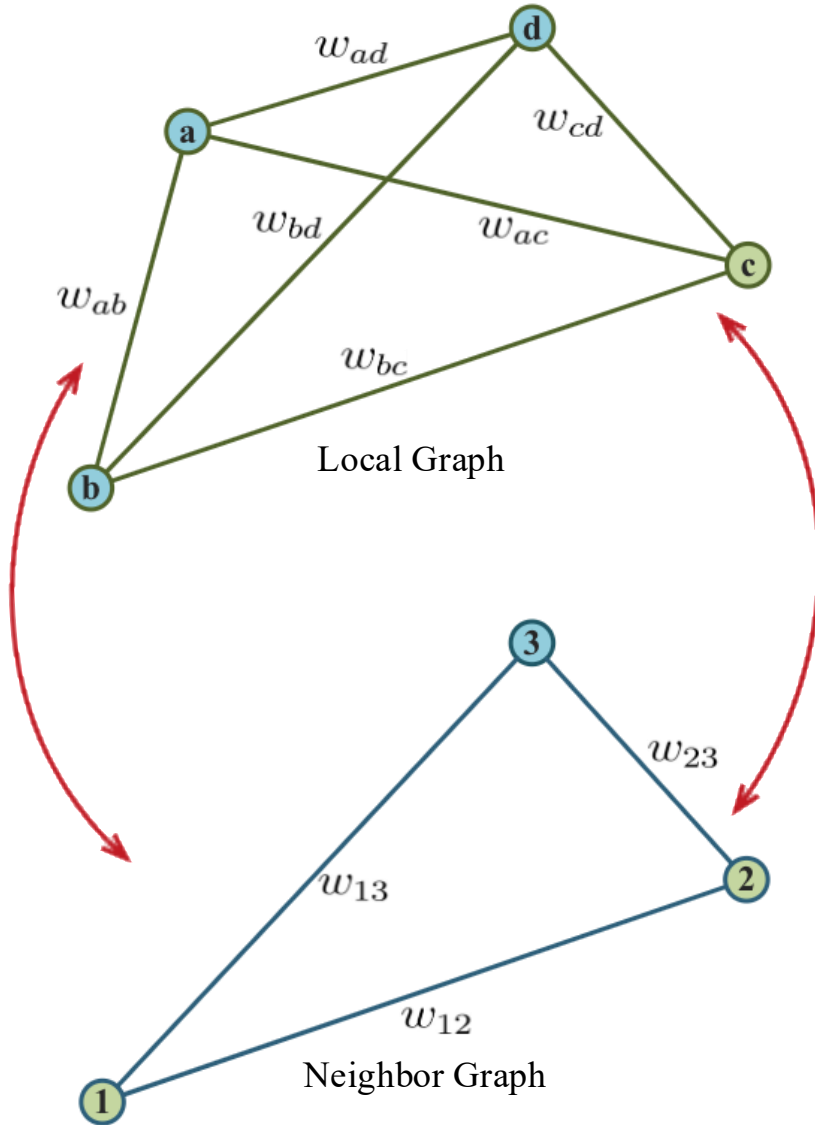
$$\forall v_j \in \mathcal{V}_l^{(t)}, \sum_{v_k \in \mathcal{V}_l^{(t-1)}} a_{jk} = 1.$$



Scene Graph of a Single Robot



Graph Matching based Inter Robot Data Association



$$A_e^* = \arg \max_{A_e} \sum_{v_j, v_k \in \mathcal{V}_l} \sum_{v_m, v_n \in \mathcal{V}_i} a_{jm} \cdot u_{jkmn} \cdot a_{kn};$$

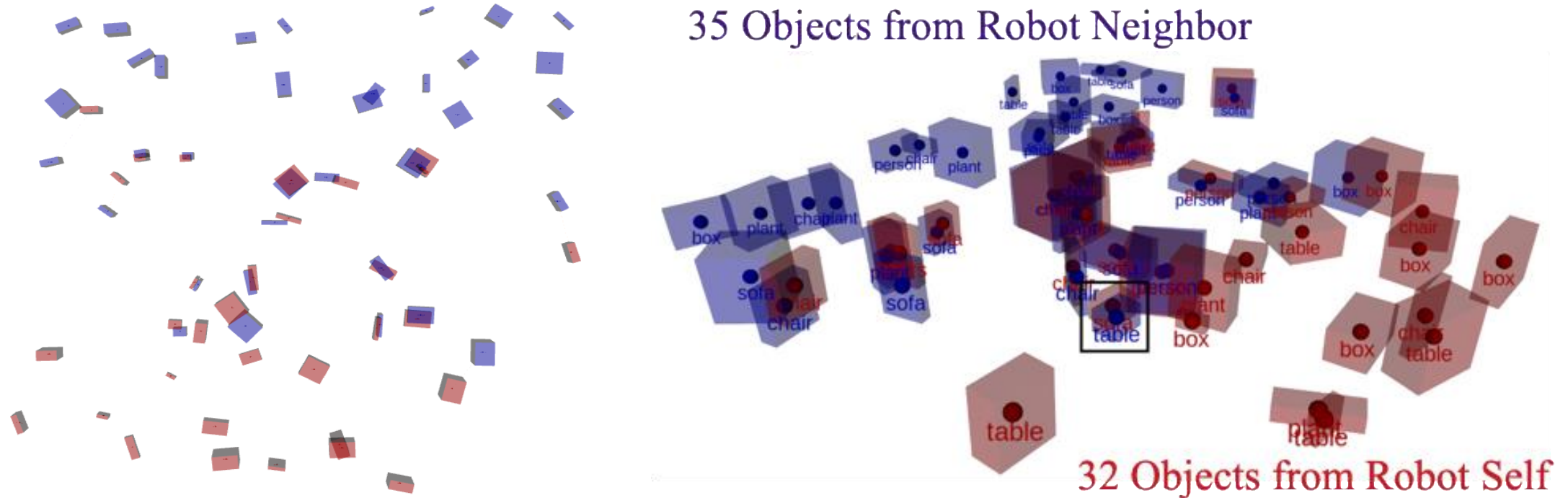
$$\forall v_k \in \mathcal{V}_i, \sum_{v_j \in \mathcal{V}_l} a_{jk} = 1;$$

$$\forall v_j \in \mathcal{V}_l, \sum_{v_k \in \mathcal{V}_i} a_{jk} = 1.$$

Vectorize into Linear Assignment Problem (LAP):

$$A_e^* = \arg \max_{A_e} \text{vec}(A_e)^\top \cdot U_{2D} \cdot \text{vec}(A_e).$$

Graph Matching Performance on Simulation Dataset



Top-down view (left) and side view (right) an example illustrating a successful transformation estimation despite the presence of both **position** and **semantic label errors**.



Scene Graph SLAM on SRI Outdoor Campus Dataset



Motivation: Underwater Multi-Robot SLAM

- **Enabling Large-Scale Marine Missions:**

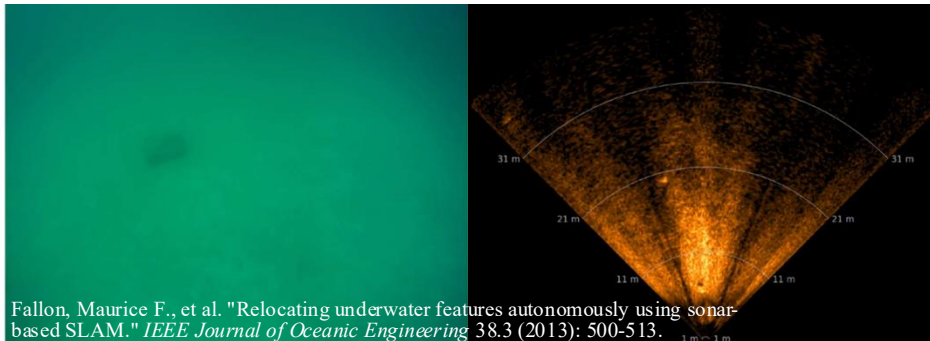
Multi-robot SLAM supports ocean exploration, offshore infrastructure maintenance, and long-term environmental monitoring.

- **Addressing Communication Constraints:**

Limited underwater **bandwidth** (e.g., 62.5 kbps acoustic modems) necessitates data-efficient multi-robot SLAM solutions.

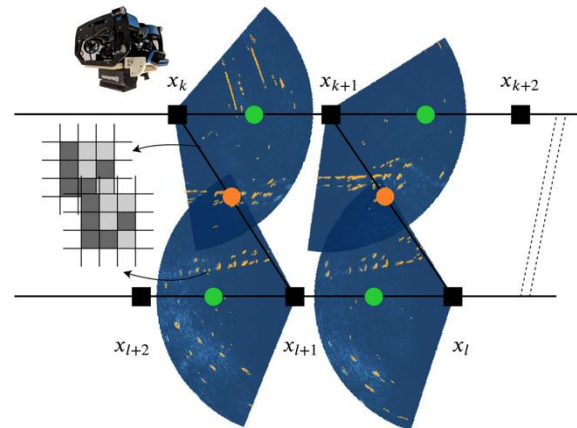
- **Enhancing Sonar-Based SLAM:**

Sonar is essential in murky waters but presents challenges in image registration due to acoustic noise and low resolution.



Fallon, Maurice F., et al. "Relocating underwater features autonomously using sonar-based SLAM." *IEEE Journal of Oceanic Engineering* 38.3 (2013): 500-513.

The visual image (left) has limited visibility, while the sonar image (right) reveals three features.



Showcase of how sonar observes landmarks in an underwater environment.



<https://bluerobotics.com/the-quest-for-edna/>
 Customized BlueROV2 for environmental data collection.

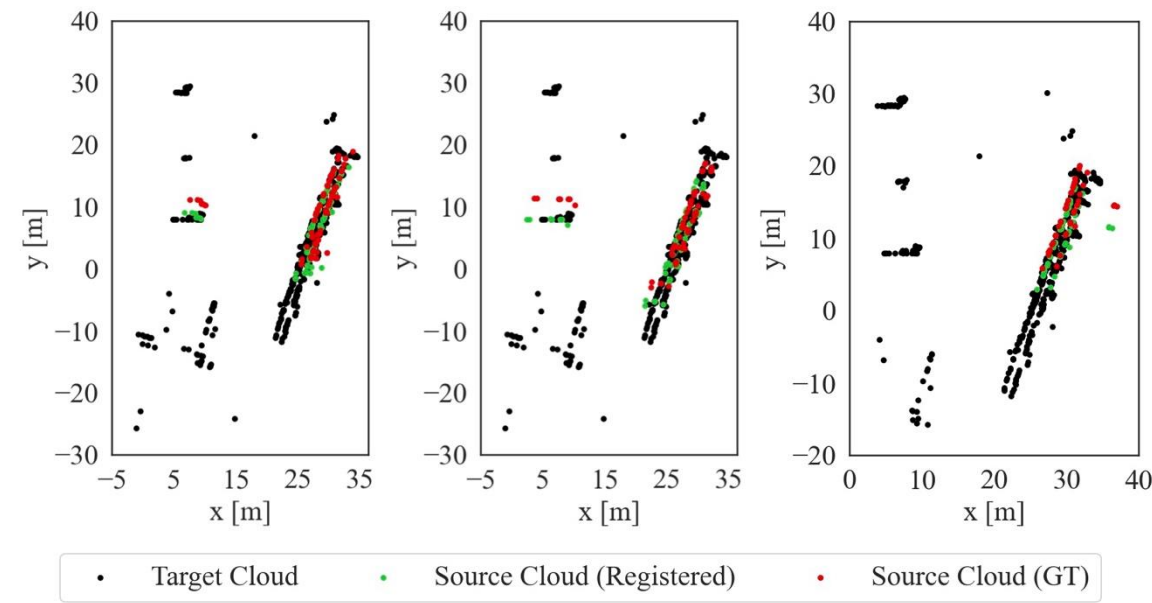


An example of an environment map built by a single robot equipped with sonar.



DRACo-SLAM2

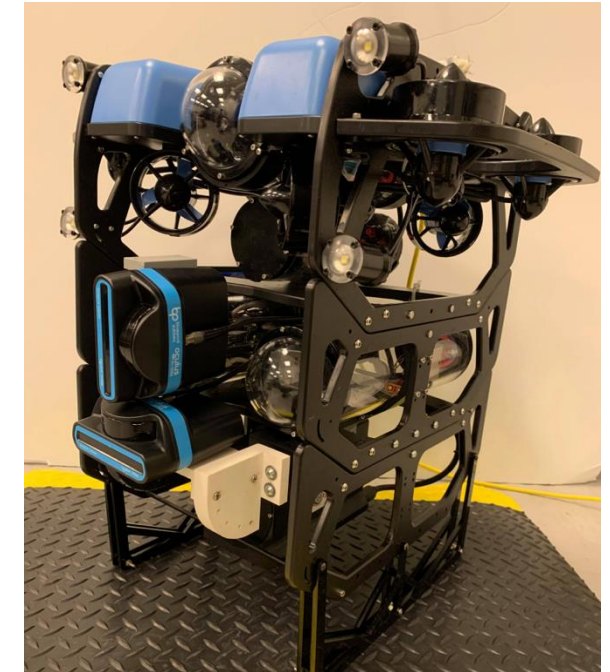
- **Object Graph Matching:** providing a **reliable initial guess** for ICP scan registration, enabling the detection of more inter-robot loop closures while maintaining **computational efficiency**.
- **Group-wise Consistent Measurement Set Maximization (GCM) Algorithm:** enhancing robustness by selecting reliable loop closures, especially in cases where multiple candidates exhibit **similar registration errors**.



An example of similar registration errors.



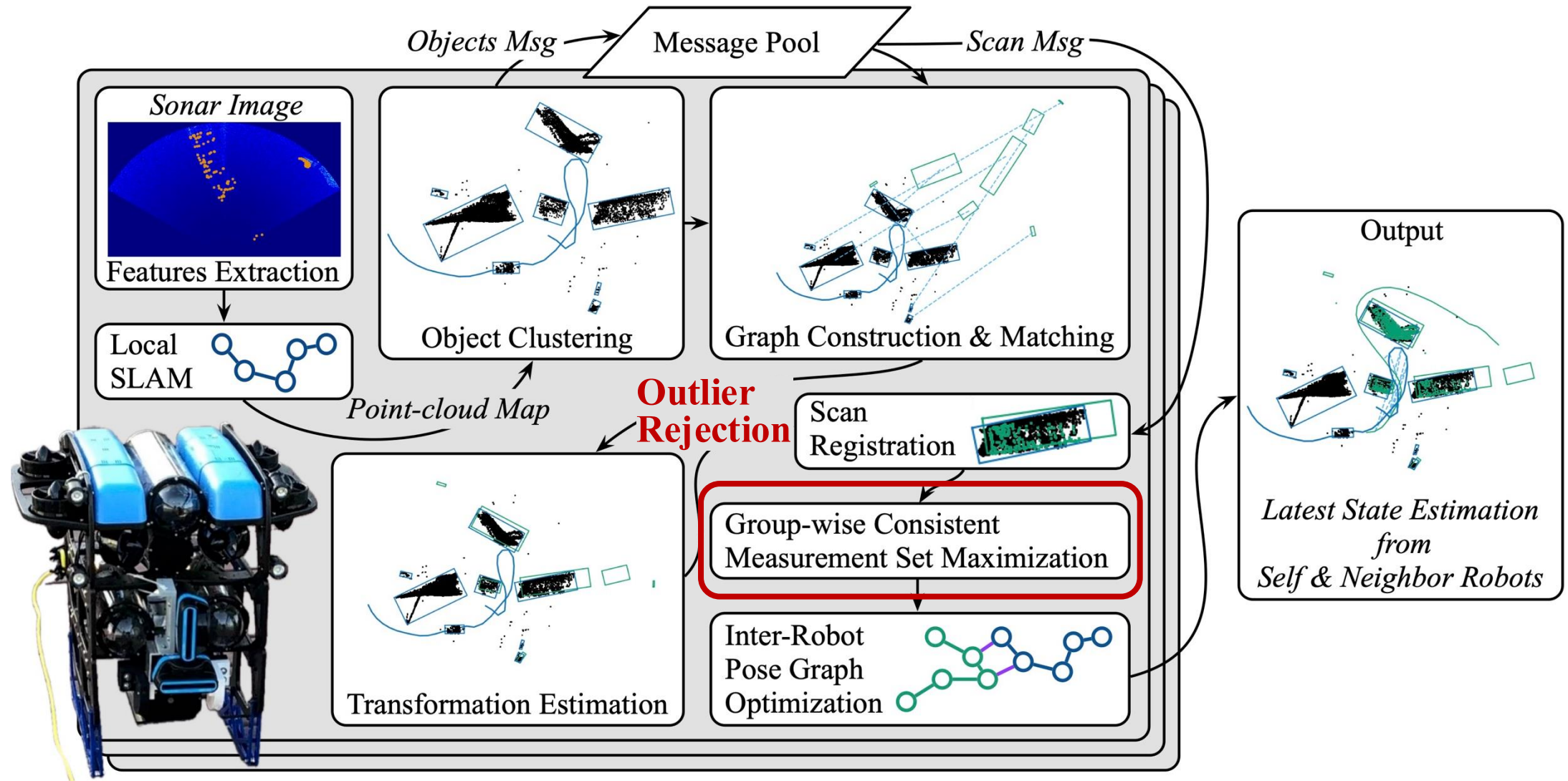
Point-cloud map and trajectory of the 3 robot USMMA real dataset.



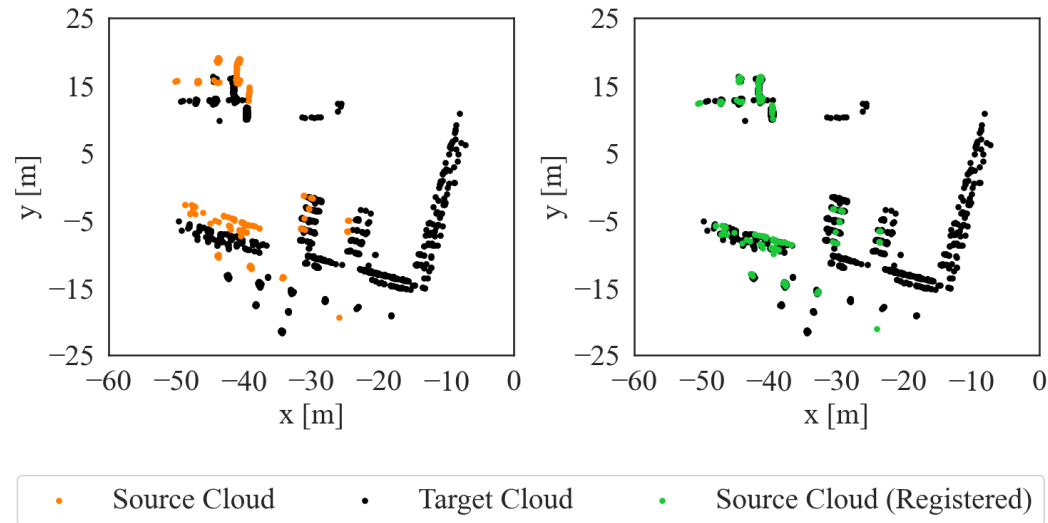
Stevens' Customized BlueROV2-Havy.



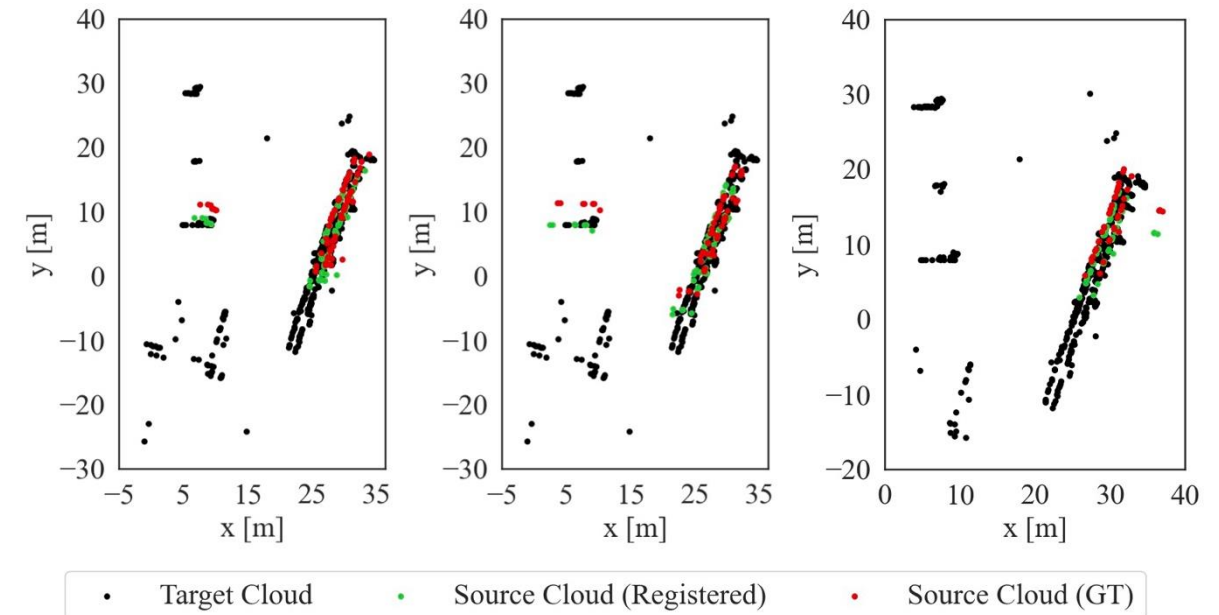
An Overview of the DRACo-SLAM2 Pipeline



Motivation: GCM



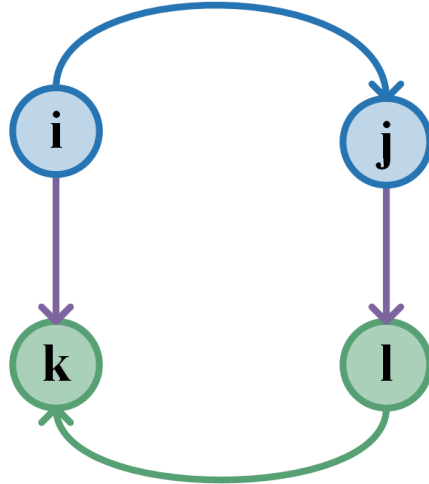
Success: The source cloud (orange) is roughly transformed into the local robot coordinate system using the transformation estimated from object graph matching.



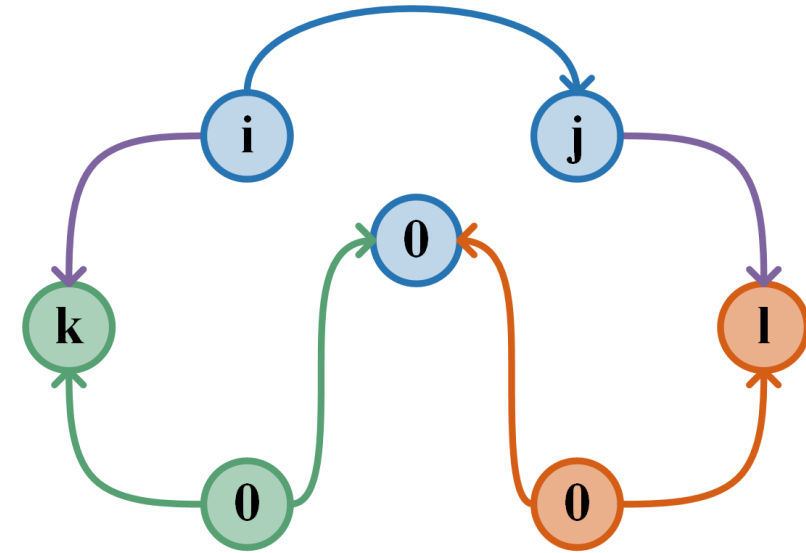
Failure: A series of three inter-robot loop closures, in the same region of the environment are impacted by **similar point cloud registration errors**.



GCM Algorithm



PCM



GCM

robot coordinate
transformation

from scene graph
matching

$$\begin{aligned}\hat{\mathbf{x}}_{\gamma_j}^{\beta_i} &= (\hat{\mathbf{x}}_{\beta_i}^{\alpha_0})^{-1} \cdot \hat{\mathbf{x}}_{\gamma_l}^{\alpha_0}, \\ &= (\hat{\mathbf{x}}_{\beta_i}^{\beta_0})^{-1} \cdot \hat{\mathbf{x}}_{\alpha_0}^{\beta_0} \cdot (\hat{\mathbf{x}}_{\alpha_0}^{\gamma_0})^{-1} \cdot \hat{\mathbf{x}}_{\gamma_j}^{\gamma_0}.\end{aligned}$$

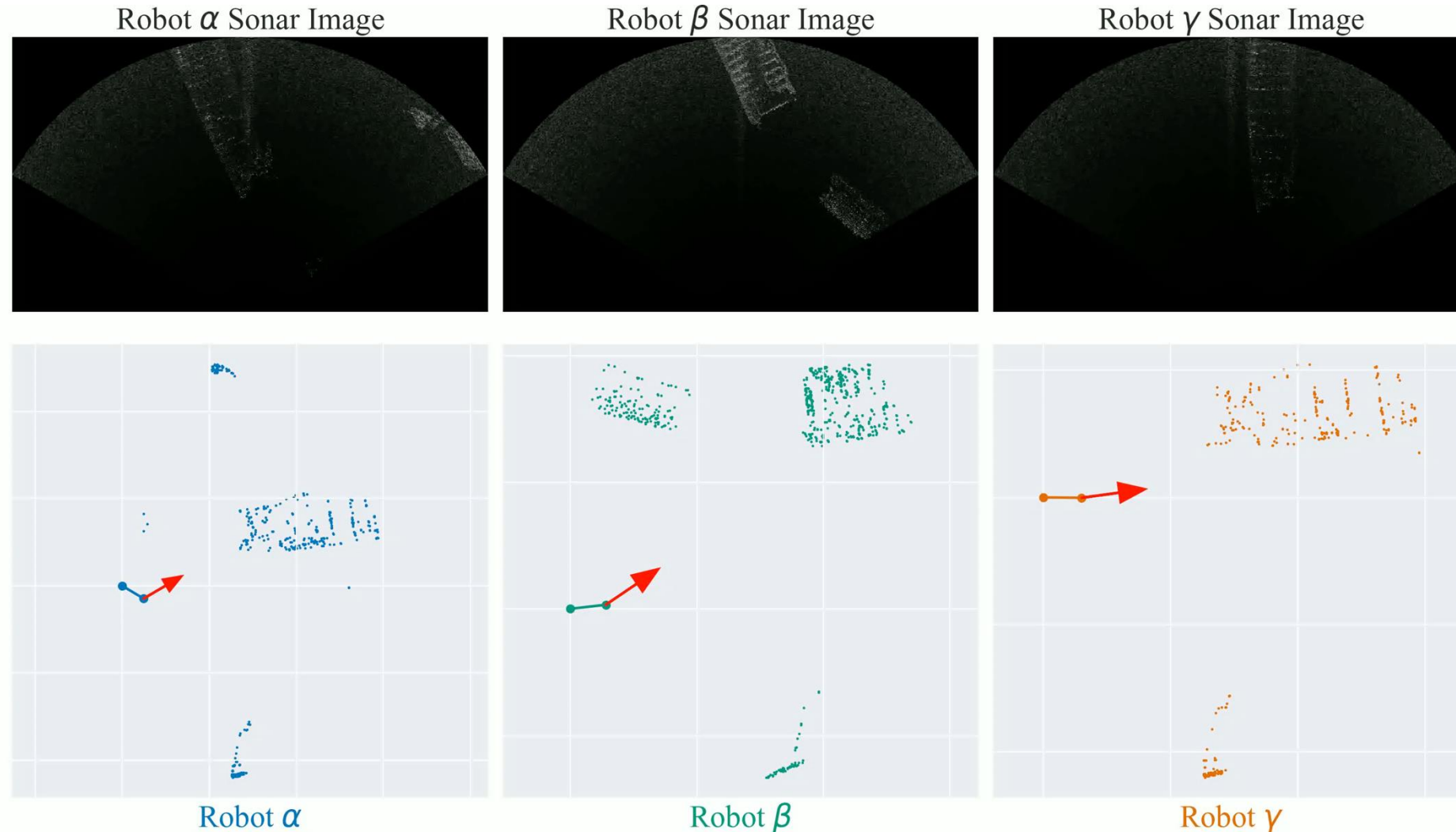
$$C(\mathbf{z}_{\alpha_k}^{\beta_i}, \mathbf{z}_{\alpha_l}^{\gamma_j}) = \left\| (\mathbf{z}_{\alpha_k}^{\beta_i})^{-1} \cdot \hat{\mathbf{x}}_{\gamma_j}^{\beta_i} \cdot \mathbf{z}_{\alpha_l}^{\gamma_j} \cdot \hat{\mathbf{x}}_{\alpha_k}^{\alpha_l} \right\|.$$

$$C(\mathbf{z}_{\alpha_k}^{\beta_i}, \mathbf{z}_{\alpha_l}^{\beta_j}) = \left\| (\mathbf{z}_{\alpha_k}^{\beta_i})^{-1} \cdot \hat{\mathbf{x}}_{\beta_j}^{\beta_i} \cdot \mathbf{z}_{\alpha_l}^{\beta_j} \cdot \hat{\mathbf{x}}_{\alpha_k}^{\alpha_l} \right\|.$$

inter-robot observation robot state estimation



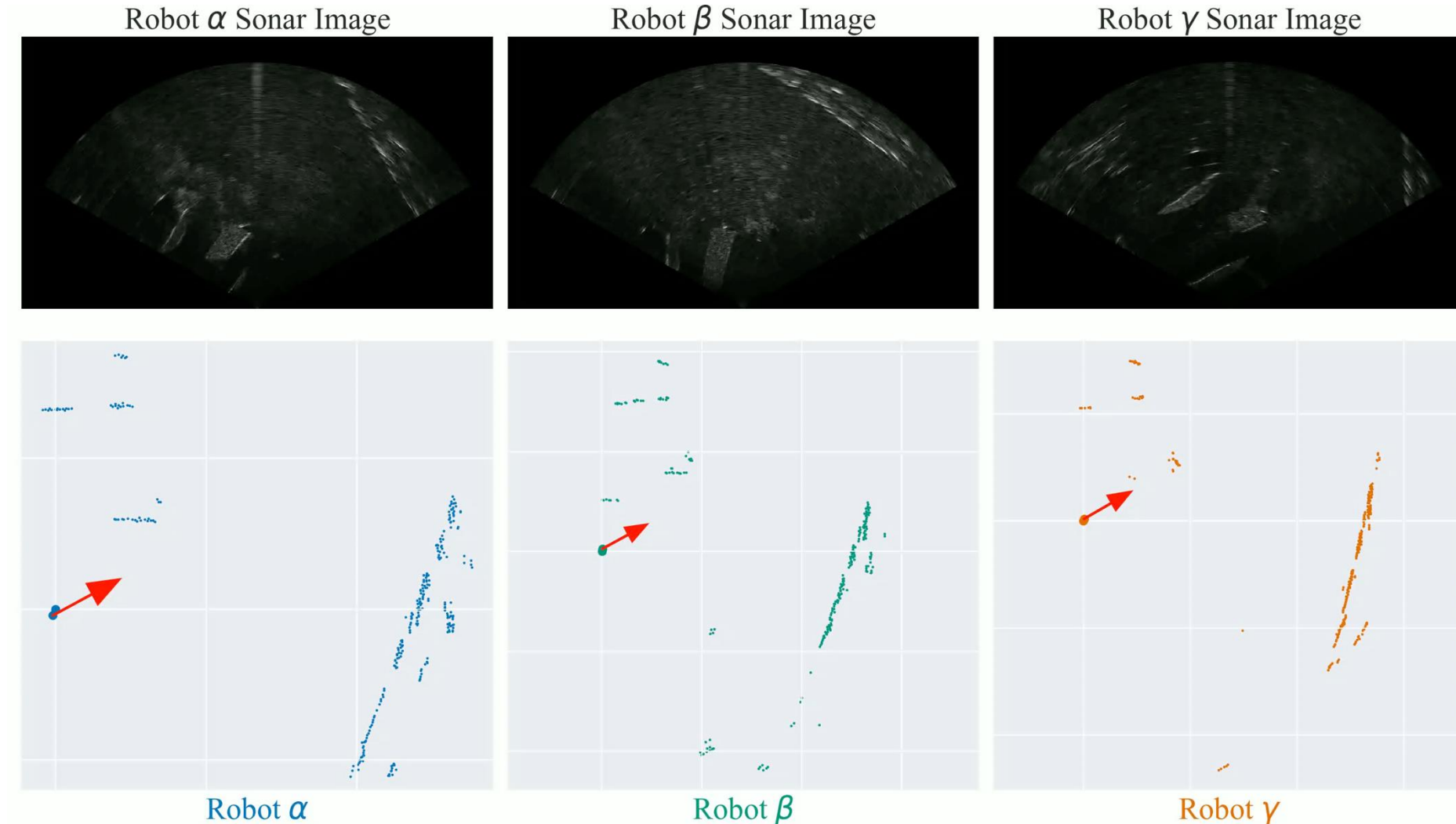
Performance on Airplane Simulation Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.



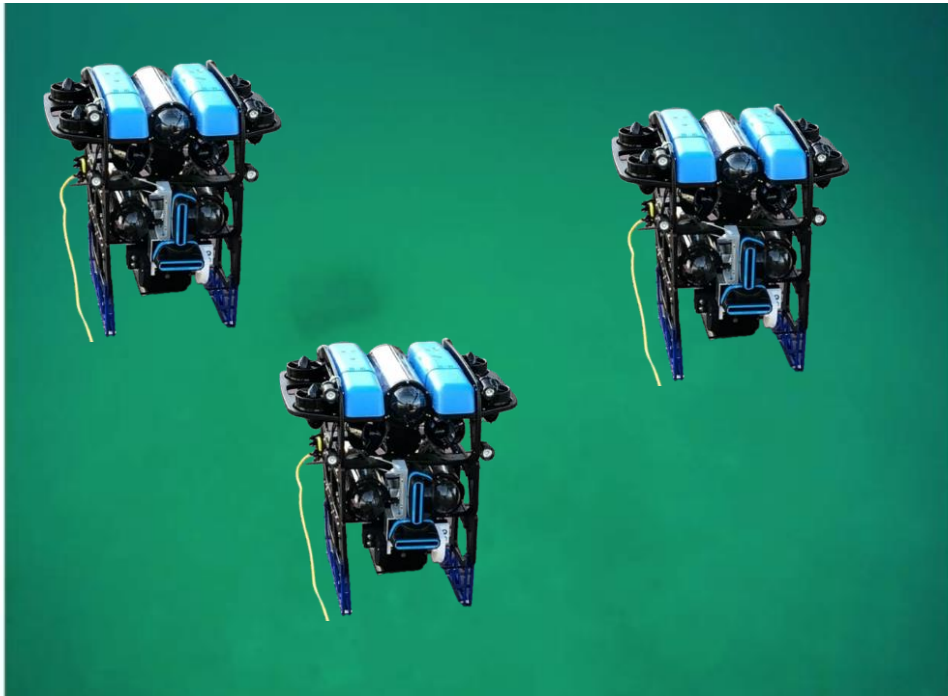
Performance on USMMA (United States Merchant Marine Academy) Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Future Work

- **Autonomous Exploration in Real Marine Environments:** Integrating DRACo-SLAM2 on underwater unmanned platforms and evaluating its performance in real-world environments.
- **Human-Robot Interactive Service:** Combining multi-robot scene graph SLAM with **large language models (LLMs)** to enable interactive motion planning and decision-making.



Fallon, Maurice F., et al. "Relocating underwater features autonomously using sonar-based SLAM." *IEEE Journal of Oceanic Engineering* 38.3 (2013): 500-513.



Screen-shoot of Big Bang Theory



Thank You!

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The Robust Field Autonomy Lab
Department of Mechanical Engineering
Stevens Institute of Technology

